

# CHETAN KUSHWAHA

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## PROFESSIONAL SUMMARY

AI/ML Engineer and Python Backend Developer with 2.5+ years of experience building production-grade AI systems across healthcare and enterprise domains. Owned end-to-end AI backends for multi-persona conversational assistants and AI-driven classification platforms — designing RAG pipelines, multi-agent orchestration with LangChain/LangGraph, and persona-based retrieval using FAISS, ChromaDB, and Azure OpenAI. Built scalable FastAPI microservices, GraphQL APIs, and secure OAuth2-based backend systems handling data at scale for clients including a global cosmetic brand's supply chain platform and an enterprise healthcare assistant serving patients, doctors, and clinical staff.

## TECHNICAL SKILLS

- **Languages:** Python
- **GenAI & LLMs:** RAG, Agentic AI, LangChain, LangGraph, OpenAI, Anthropic, Gemini, Llama, NLTK, Sentence-Transformers, Advanced Prompt Engineering, Few-Shot Prompting, Chain-of-Thought, Function Calling, Structured Outputs, Vector Search, Embeddings, MCP, HuggingFace, Semantic Kernel, Fine-Tuning, Multi-Agent Systems, AI Agents, Prompt Flow, Context Engineering
- **Backend & APIs:** FastAPI, Flask, REST APIs, GraphQL, WebSockets, OAuth2.0, AsyncIO, Pydantic, SQLAlchemy, Alembic, Background Tasks, API Versioning, Swagger/OpenAPI, Dependency Injection, JWT, RBAC, Microservices, System Design, Git, GitHub, CI/CD, GitHub Actions
- **Cloud & Platforms:** AWS Bedrock, SageMaker, Azure OpenAI, Vertex AI, Azure AI Studio, Azure AI Foundry, Azure Machine Learning, Azure Cognitive Services, Azure AI Document Intelligence, Azure Blob Storage, Azure DevOps, Docker, Redis, Celery, Linux/Unix
- **Data & ML Tooling:** FAISS, ChromaDB, Milvus, Pinecone, Azure AI Search, Hybrid Search, Semantic Search, Neo4j, MySQL, PostgreSQL, MongoDB, NumPy, Pandas, Prometheus, Grafana

## PROFESSIONAL EXPERIENCE

**Python Backend Developer** | Sutherland Global Inc. – ELC EHSAI

*Dec 2025 – Present*

- Owned and built the AI-driven classification engine that extracts and parses Safety Data Sheets (SDS) using Azure AI Document Intelligence, automatically suggesting Dangerous Goods (DG) and GHS hazard classifications for raw materials sourced from tens of thousands of vendor records.
- Designed and maintained the core Python backend service powering the platform, exposing classification data to the frontend through GraphQL APIs built with FastAPI and Strawberry.
- Built a human-in-the-loop review workflow enabling the EHS (Environment, Health & Safety) team to approve or reject AI-suggested classifications, replacing a fully manual raw-material classification process for materials from multiple vendors.
- Implemented writeback logic to sync approved classifications back to the client's source systems — RDS for GHS hazard classification and SCS for DG/supply chain shipping classification — ensuring consistent downstream data across enterprise systems.
- Worked with Oracle Database for persistence and query optimization, supporting classification and writeback operations at scale.

**AI/ML Engineer** | Sutherland Global Inc. – Mikaila Platform

*Mar 2024 – Dec 2025*

- Owned the end-to-end AI backend for Mikaila, a production-grade multi-persona enterprise GenAI assistant serving patients, doctors, and clinical support staff on a single platform, built on Azure OpenAI-compatible LLMs and NVIDIA AI technologies.
- Designed an orchestrator agent using LangChain and LangGraph that routes queries based on user persona and intent — directing patient symptom/disease questions and doctor/staff research and knowledge-base queries through tailored retrieval and response flows.
- Built persona-aware Retrieval-Augmented Generation (RAG) pipelines with FAISS, ChromaDB, and Pinecone, filtering knowledge-base access by persona ID so patients, doctors, and staff each retrieve context relevant and appropriate to their role.
- Contributed to a 40% reduction in oncology research time for clinical staff by architecting the semantic retrieval layer connecting research papers and internal knowledge sources to the assistant.
- Integrated sentiment analysis into the patient-facing conversational flow to generate more empathetic responses, improving patient experience when discussing symptoms and health concerns.
- Developed and maintained parts of the backend API layer using FastAPI, asynchronous Python, OAuth2, and Docker, supporting secure multi-persona access and deployment of the AI backend.

## CERTIFICATIONS

Deep Learning Specialization – DeepLearning.AI | Introduction to LangGraph – LangChain Academy

## EDUCATION

Diploma in Advance Computing, Sunbeam Infotech, Pune  
B.Tech, Computer Science, Shri Ram Institute of Technology, Jabalpur

*Sep 2023 – Mar 2024*  
*Aug 2019 – May 2023*